

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc. I.T Semester V)
Applied Business Analytics
M2 Practical – XII

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Class: TYIT	Batch: 1
Date of Assignment: 03-10-26	Date/Time of Submission: 03-10-26

12. Analysis of Variance and Experimental Design:

a. Design and conduct an experiment to study the effects of different factors on a specific response variable, analyze the results using analysis of variance (ANOVA), and draw conclusions.

Code:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import f_oneway
import statsmodels.api as sm
from statsmodels.formula.api import ols

# LOAD DATASET
file_path = "Retail and warehouse Sale.csv"
df = pd.read_csv(file_path)

print("Dataset loaded successfully!")
# PART A – ANALYSIS OF VARIANCE (ANOVA)
# 1. Display first five records
print("\n1. First Five Records:")
print(df.head())

# 2. Number of rows and columns
print("\n2. Dataset Shape:")
print("Number of Rows:", df.shape[0])
print("Number of Columns:", df.shape[1])

# 3. Column names and data types
print("\n3. Column Names and Data Types:")
print(df.dtypes)

# 4. Missing values and duplicate records
print("\n4. Missing Values:")
print(df.isnull().sum())

print("\nNumber of Duplicate Records:")
print(df.duplicated().sum())
```

```

# 5. Different ITEM TYPE categories
print("\n5. Different ITEM TYPE Categories:")
print(df["ITEM TYPE"].unique())

print("\nNumber of ITEM TYPE Categories:")
print(df["ITEM TYPE"].nunique())

# 6. Average RETAIL SALES for each ITEM TYPE
average_sales = df.groupby("ITEM TYPE")["RETAIL SALES"].mean()

print("\n6. Average RETAIL SALES for Each ITEM TYPE:")
print(average_sales)

# 7. Box plot
plt.figure(figsize=(12, 6))
sns.boxplot(
    x="ITEM TYPE",
    y="RETAIL SALES",
    data=df
)

plt.title("Retail Sales by Item Type")
plt.xlabel("Item Type")
plt.ylabel("Retail Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# 8. Hypotheses
print("\n8. Hypotheses:")
print("H0: The average RETAIL SALES are equal for all ITEM TYPE categories.")
print("H1: At least one ITEM TYPE has a different average RETAIL SALES.")

# 9 & 10. One-way ANOVA
groups = [
    group["RETAIL SALES"].dropna()
    for name, group in df.groupby("ITEM TYPE")
]

f_statistic, p_value = f_oneway(*groups)

print("\n9 & 10. One-Way ANOVA Results:")
print("F-statistic:", f_statistic)
print("p-value:", p_value)

# 11 & 12. Compare p-value with 0.05
alpha = 0.05

print("\n11 & 12. Hypothesis Test:")

```

```

if p_value < alpha:
    print("p-value < 0.05")
    print("Reject the Null Hypothesis (H0).")
else:
    print("p-value >= 0.05")
    print("Do Not Reject the Null Hypothesis (H0).")

# 13. Conclusion
print("\n13. Conclusion:")

if p_value < alpha:
    print(
        "There is a statistically significant difference in average "
        "RETAIL SALES among the different ITEM TYPE categories."
    )
else:
    print(
        "There is no statistically significant difference in average "
        "RETAIL SALES among the different ITEM TYPE categories."
    )

```

Output:

Dataset loaded successfully!

1. First Five Records:

	YEAR	MONTH	SUPPLIER	ITEM CODE	\
0	2020	1	REPUBLIC NATIONAL DISTRIBUTING CO	100009	
1	2020	1	PWSMN INC	100024	
2	2020	1	RELIABLE CHURCHILL LLLP	1001	
3	2020	1	LANTERNA DISTRIBUTORS INC	100145	
4	2020	1	DIONYSOS IMPORTS INC	100293	

	ITEM DESCRIPTION	ITEM TYPE	RETAIL SALES	\
0	BOOTLEG RED - 750ML	WINE	0.00	
1	MOMENT DE PLAISIR - 750ML	WINE	0.00	
2	S SMITH ORGANIC PEAR CIDER - 18.70Z	BEER	0.00	
3	SCHLINK HAUS KABINETT - 750ML	WINE	0.00	
4	SANTORINI GAVALA WHITE - 750ML	WINE	0.82	

	RETAIL TRANSFERS	WAREHOUSE SALES
0	0.0	2.0
1	1.0	4.0
2	0.0	1.0
3	0.0	1.0
4	0.0	0.0

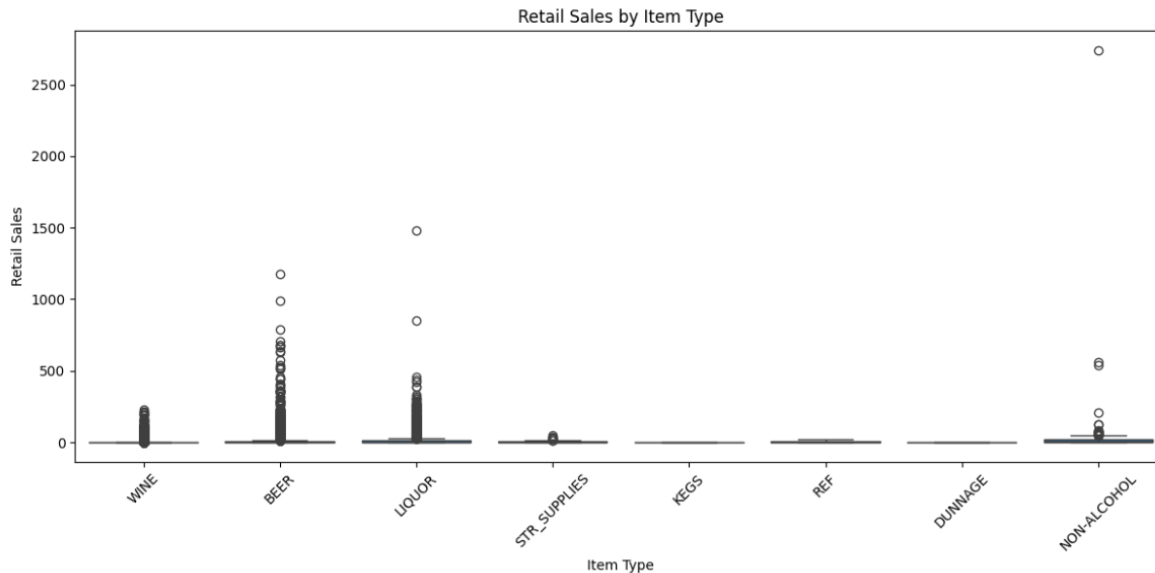
2. Dataset Shape:

```

...
REF          5.783750
STR_SUPPLIES 5.485714
WINE         3.195334
Name: RETAIL SALES, dtype: float64

```

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8. Hypotheses:

H0: The average RETAIL SALES are equal for all ITEM TYPE categories.

H1: At least one ITEM TYPE has a different average RETAIL SALES.

9 & 10. One-Way ANOVA Results:

F-statistic: 123.52155338346161

p-value: 8.685629095380671e-180

11 & 12. Hypothesis Test:

p-value < 0.05

Reject the Null Hypothesis (H0).

13. Conclusion:

There is a statistically significant difference in average RETAIL SALES among the different ITEM TYPE categories.

b. Implement a factorial experiment and analyze the main effects and interaction effects of factors using statistical techniques.

Code:

```
print("\n\n=====")
print("PART B – FACTORIAL EXPERIMENT")
print("=====")
```

```
# 14. Select ITEM TYPE and YEAR as factors
```

```
print("\n14. Factors:")
print("Factor 1: ITEM TYPE")
print("Factor 2: YEAR")
print("Response Variable: RETAIL SALES")
```

```
# Check available years
```

```
print("\nAvailable Years:")
print(sorted(df["YEAR"].dropna().unique()))
```

```
print("\nNumber of Different Years:",
      df["YEAR"].nunique())
```

```
# 15. Average retail sales for each combination
```

```

combination_average = (
    df.groupby(["ITEM TYPE", "YEAR"])["RETAIL SALES"]
      .mean()
      .reset_index()
)

print("\n15. Average RETAIL SALES for Each ITEM TYPE and YEAR:")
print(combination_average)

# Pivot table for better display
pivot_table = pd.pivot_table(
    df,
    values="RETAIL SALES",
    index="ITEM TYPE",
    columns="YEAR",
    aggfunc="mean"
)

print("\nAverage Sales Pivot Table:")
print(pivot_table)

# 16. Visualization
plt.figure(figsize=(12, 6))

sns.barplot(
    data=df,
    x="ITEM TYPE",
    y="RETAIL SALES",
    hue="YEAR",
    estimator="mean",
    errorbar=None
)

plt.title("Average Retail Sales by Item Type and Year")
plt.xlabel("Item Type")
plt.ylabel("Average Retail Sales")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()

# 17. Main effect of ITEM TYPE
print("\n17. Main Effect of ITEM TYPE:")

item_type_means = (
    df.groupby("ITEM TYPE")["RETAIL SALES"]
      .mean()
      .sort_values(ascending=False)
)

```

```

print(item_type_means)

# 18. Main effect of YEAR
print("\n18. Main Effect of YEAR:")
year_means = (
    df.groupby("YEAR")["RETAIL SALES"]
    .mean()
)

print(year_means)

# 19–22. Factorial ANOVA
print("\n19–22. Factorial ANOVA:")
if df["YEAR"].nunique() < 2:

    print("\nWARNING:")
    print("The dataset contains only one YEAR:",
          df["YEAR"].unique()[0])

    print(
        "\nFactorial ANOVA cannot be performed using YEAR "
        "because at least two different YEAR levels are required."
    )

    print("\nTherefore:")
    print("ITEM TYPE main effect: Can be analyzed.")
    print("YEAR main effect: Cannot be tested statistically.")
    print("ITEM TYPE × YEAR interaction: Cannot be tested.")

else:

    # Two-way factorial ANOVA
    model = ols(
        'Q("RETAIL SALES") ~ C(Q("ITEM TYPE")) * C(YEAR)',
        data=df
    ).fit()

    anova_table = sm.stats.anova_lm(model, typ=2)

    print("\n20. Factorial ANOVA Results:")
    print(anova_table)

    print("\n21. P-values:")

    for effect in anova_table.index:
        print(effect, "p-value =",
              anova_table.loc[effect, "PR(>F)"])

```

```

# 22. Identify significant effects
print("\n22. Significance at alpha = 0.05:")

for effect in anova_table.index:

    if effect == "Residual":
        continue

    p = anova_table.loc[effect, "PR(>F)"]

    if p < 0.05:
        print(effect, "-> Significant")
    else:
        print(effect, "-> Not Significant")

# 23. Final Business Conclusion
print("\n23. FINAL BUSINESS CONCLUSION:")
if p_value < 0.05:
    print(
        "The one-way ANOVA indicates that ITEM TYPE has a "
        "significant effect on RETAIL SALES. Therefore, "
        "retail sales vary significantly among different "
        "item categories."
    )
else:
    print(
        "The one-way ANOVA indicates that ITEM TYPE does not "
        "have a statistically significant effect on RETAIL SALES."
    )

if df["YEAR"].nunique() < 2:
    print(
        "The uploaded dataset contains only the year 2020. "
        "Hence, the effect of YEAR and the ITEM TYPE × YEAR "
        "interaction cannot be statistically evaluated."
    )
else:
    print(
        "The factorial ANOVA evaluates the effects of ITEM TYPE, "
        "YEAR, and their interaction on RETAIL SALES."
    )

```

Output:

```
=====
PART B - FACTORIAL EXPERIMENT
=====
```

```
14. Factors:
Factor 1: ITEM TYPE
Factor 2: YEAR
Response Variable: RETAIL SALES
```

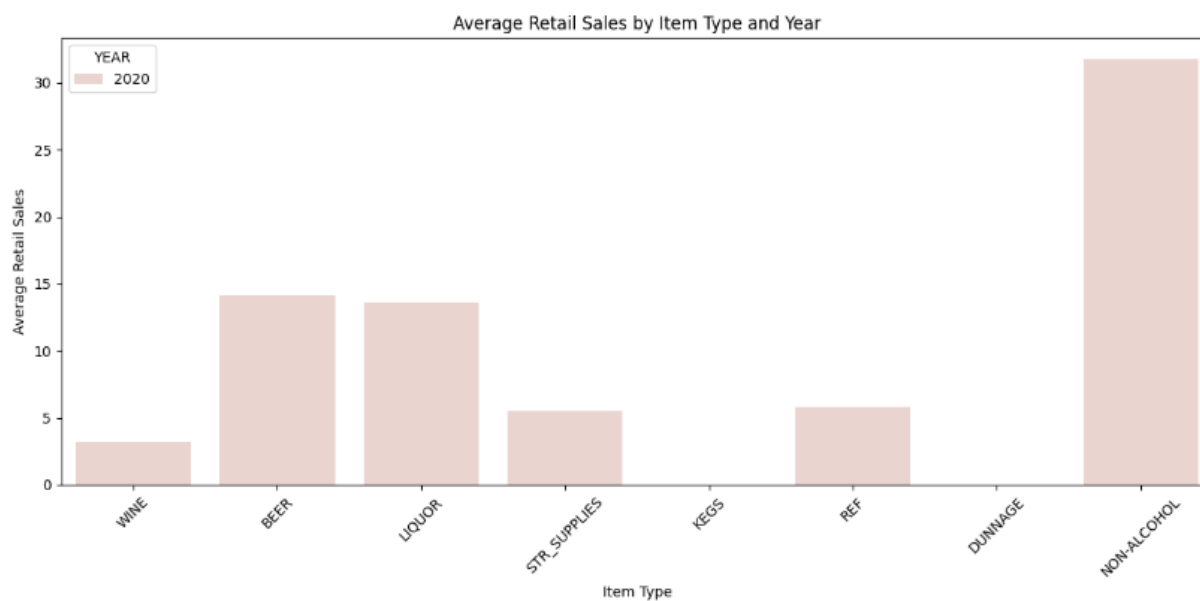
```
Available Years:
[np.int64(2020)]
```

```
Number of Different Years: 1
```

```
15. Average RETAIL SALES for Each ITEM TYPE and YEAR:
```

	ITEM TYPE	YEAR	RETAIL SALES
0	BEER	2020	14.118748
1	DUNNAGE	2020	0.000000
2	KEGS	2020	0.000000
3	LIQUOR	2020	13.635171
4	NON-ALCOHOL	2020	31.742419
5	REF	2020	5.783750
6	STR_SUPPLIES	2020	5.485714
...			
	NON-ALCOHOL		31.742419
	REF		5.783750
	STR_SUPPLIES		5.485714
	WINE		3.195334

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```
17. Main Effect of ITEM TYPE:
ITEM TYPE
NON-ALCOHOL    31.742419
BEER           14.118748
LIQUOR         13.635171
REF            5.783750
STR_SUPPLIES   5.485714
WINE           3.195334
KEGS           0.000000
DUNNAGE        0.000000
Name: RETAIL SALES, dtype: float64
```

```
18. Main Effect of YEAR:
YEAR
2020    6.939796
Name: RETAIL SALES, dtype: float64
```

```
19-22. Factorial ANOVA:
```

```
WARNING:
```

```
The dataset contains only one YEAR: 2020
```

```
Factorial ANOVA cannot be performed using YEAR because at least two different YEAR levels are required.
...
```

CONCLUSION:

The one-way ANOVA indicates that ITEM TYPE has a significant effect on RETAIL SALES.

Therefore, retail sales vary significantly among different item categories.

The uploaded dataset contains only the year 2020. Hence, the effect of YEAR and the ITEM TYPE $\times$ YEAR interaction cannot be statistically evaluated.